# 1. Persiapan Data

## Import library yang diperlukan

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import time
from itertools import product
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.model_selection import ParameterGrid
from imblearn.under_sampling import RandomUnderSampler
from imblearn.over_sampling import RandomOverSampler, SMOTE
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, classification_report, confusion_matrix
from google.colab import drive
```

## Sambungkan ke Google Drive dan arahkan ke path data yang akan digunakan

```
drive.mount('/content/drive')
```

```
path_data = "/content/drive/MyDrive/Bengkel Coding/Pertemuan 3/Dataset/adult.csv"
```

```
Mounted at /content/drive
```

## Baca dataset

Pada data yang kita gunakan kali ini, tantangannya adalah bahwa data ini masih belum memiliki nama kolom, maka nanti akan kita beri nama pada kolomnya termasuk akan kita analisis mengenai langkah apa saja yang akan kita gunakan dalam melakukan pre-processing data

```
df = pd.read_csv(path_data)
df.head()
```

| | 39 | State-gov | 77516 | Bachelors | 13 | Never-married | Adm-clerical | Not-in-family | White | Male | 2174 | 0 | 40 | United-States | <=50K |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 50 | Self-emp-not-inc | 83311 | Bachelors | 13 | Married-civ-spouse | Exec-managerial | Husband | White | Male | 0 | 0 | 13 | United-States | <=50K |
| **1** | 38 | Private | 215646 | HS-grad | 9 | Divorced | Handlers-cleaners | Not-in-family | White | Male | 0 | 0 | 40 | United-States | <=50K |
| **2** | 53 | Private | 234721 | 11th | 7 | Married-civ-spouse | Handlers-cleaners | Husband | Black | Male | 0 | 0 | 40 | United-States | <=50K |
| **3** | 28 | Private | 338409 | Bachelors | 13 | Married-civ-spouse | Prof-specialty | Wife | Black | Female | 0 | 0 | 40 | Cuba | <=50K |
| **4** | 37 | Private | 284582 | Masters | 14 | Married-civ-spouse | Exec-managerial | Wife | White | Female | 0 | 0 | 40 | United-States | <=50K |

Langkah berikutnya: Buat kode dengan df | Lihat plot yang direkomendasikan | New interactive sheet

```
df.tail()
```

| | 39 | State-gov | 77516 | Bachelors | 13 | Never-married | Adm-clerical | Not-in-family | White | Male | 2174 | 0 | 40 | United-States | <=50K |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **32555** | 27 | Private | 257302 | Assoc-acdm | 12 | Married-civ-spouse | Tech-support | Wife | White | Female | 0 | 0 | 38 | United-States | <=50K |
| **32556** | 40 | Private | 154374 | HS-grad | 9 | Married-civ-spouse | Machine-op-inspct | Husband | White | Male | 0 | 0 | 40 | United-States | >50K |
| **32557** | 58 | Private | 151910 | HS-grad | 9 | Widowed | Adm-clerical | Unmarried | White | Female | 0 | 0 | 40 | United-States | <=50K |
| **32558** | 22 | Private | 201490 | HS-grad | 9 | Never-married | Adm-clerical | Own-child | White | Male | 0 | 0 | 20 | United-States | <=50K |
| **32559** | 52 | Self-emp-inc | 287927 | HS-grad | 9 | Married-civ-spouse | Exec-managerial | Wife | White | Female | 15024 | 0 | 40 | United-States | >50K |

# 2. Pre-Processing Data

## Memberikan nama kolom pada data

```
col_names = ['age', 'workclass', 'fnlwgt', 'education', 'education_num', 'marital_status',
             'occupation', 'relationship', 'race', 'sex', 'capital_gain', 'capital_loss',
             'hours_per_week', 'native_country', 'income']
```

```
df.columns = col_names
df.head()
```

| | age | workclass | fnlwgt | education | education_num | marital_status | occupation | relationship | race | sex | capital_gain | capital_loss | hours_per_week |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 50 | Self-emp-not-inc | 83311 | Bachelors | 13 | Married-civ-spouse | Exec-managerial | Husband | White | Male | 0 | 0 | 13 |
| **1** | 38 | Private | 215646 | HS-grad | 9 | Divorced | Handlers-cleaners | Not-in-family | White | Male | 0 | 0 | 40 |
| **2** | 53 | Private | 234721 | 11th | 7 | Married-civ-spouse | Handlers-cleaners | Husband | Black | Male | 0 | 0 | 40 |
| **3** | 28 | Private | 338409 | Bachelors | 13 | Married-civ-spouse | Prof-specialty | Wife | Black | Female | 0 | 0 | 40 |
| **4** | 37 | Private | 284582 | Masters | 14 | Married-civ-spouse | Exec-managerial | Wife | White | Female | 0 | 0 | 40 |

Langkah berikutnya: Buat kode dengan df | Lihat plot yang direkomendasikan | New interactive sheet

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32560 entries, 0 to 32559
Data columns (total 15 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   age             32560 non-null  int64
 1   workclass       32560 non-null  object
 2   fnlwgt          32560 non-null  int64
 3   education       32560 non-null  object
```

```
 4   education_num   32560 non-null  int64
 5   marital_status  32560 non-null  object
 6   occupation      32560 non-null  object
 7   relationship    32560 non-null  object
 8   race            32560 non-null  object
 9   sex             32560 non-null  object
 10  capital_gain    32560 non-null  int64
 11  capital_loss    32560 non-null  int64
 12  hours_per_week  32560 non-null  int64
 13  native_country  32560 non-null  object
 14  income          32560 non-null  object
dtypes: int64(6), object(9)
memory usage: 3.7+ MB
```

tidak ditemukan data yang hilang karena jumlah seluruh value sama.

```
df.describe()
```

| | age | fnlwgt | education_num | capital_gain | capital_loss | hours_per_week |
|---|---|---|---|---|---|---|
| **count** | 32560.000000 | 3.256000e+04 | 32560.000000 | 32560.000000 | 32560.000000 | 32560.000000 |
| **mean** | 38.581634 | 1.897818e+05 | 10.080590 | 1077.615172 | 87.306511 | 40.437469 |
| **std** | 13.640642 | 1.055498e+05 | 2.572709 | 7385.402999 | 402.966116 | 12.347618 |
| **min** | 17.000000 | 1.228500e+04 | 1.000000 | 0.000000 | 0.000000 | 1.000000 |
| **25%** | 28.000000 | 1.178315e+05 | 9.000000 | 0.000000 | 0.000000 | 40.000000 |
| **50%** | 37.000000 | 1.783630e+05 | 10.000000 | 0.000000 | 0.000000 | 40.000000 |
| **75%** | 48.000000 | 2.370545e+05 | 12.000000 | 0.000000 | 0.000000 | 45.000000 |
| **max** | 90.000000 | 1.484705e+06 | 16.000000 | 99999.000000 | 4356.000000 | 99.000000 |

## Menghitung nilai unik pada masing-masing class

Di sini kita akan melakukan pengecekan terkait nilai-nilai unik yang ada pada tiap-tiap kolom yang bertipe kategorikal

```
unique_counts = df.nunique()
print(unique_counts)
```

```
age                  73
workclass             9
fnlwgt            21647
education            16
education_num        16
marital_status        7
occupation           15
relationship          6
race                  5
sex                   2
capital_gain        119
capital_loss         92
hours_per_week       94
native_country       42
income                2
dtype: int64
```

di sini kita melakukan pengecekan value terhadap seluruh kolom yang berisi string

```
kolom = ["workclass", "education", "marital_status",    "occupation",   "relationship", "race", "sex", "native_country",    "income"]
for col in kolom:
    print(f"Kolom: {col}")
    print(df[col].unique())
    print("-" * 40)
```

```
Kolom: workclass
[' Self-emp-not-inc' ' Private' ' State-gov' ' Federal-gov' ' Local-gov'
 ' ?' ' Self-emp-inc' ' Without-pay' ' Never-worked']
----------------------------------------
Kolom: education
[' Bachelors' ' HS-grad' ' 11th' ' Masters' ' 9th' ' Some-college'
 ' Assoc-acdm' ' Assoc-voc' ' 7th-8th' ' Doctorate' ' Prof-school'
 ' 5th-6th' ' 10th' ' 1st-4th' ' Preschool' ' 12th']
----------------------------------------
Kolom: marital_status
[' Married-civ-spouse' ' Divorced' ' Married-spouse-absent'
 ' Never-married' ' Separated' ' Married-AF-spouse' ' Widowed']
----------------------------------------
Kolom: occupation
[' Exec-managerial' ' Handlers-cleaners' ' Prof-specialty'
 ' Other-service' ' Adm-clerical' ' Sales' ' Craft-repair'
 ' Transport-moving' ' Farming-fishing' ' Machine-op-inspct'
 ' Tech-support' ' ?' ' Protective-serv' ' Armed-Forces'
 ' Priv-house-serv']
----------------------------------------
Kolom: relationship
[' Husband' ' Not-in-family' ' Wife' ' Own-child' ' Unmarried'
 ' Other-relative']
----------------------------------------
Kolom: race
[' White' ' Black' ' Asian-Pac-Islander' ' Amer-Indian-Eskimo' ' Other']
----------------------------------------
Kolom: sex
[' Male' ' Female']
----------------------------------------
Kolom: native_country
[' United-States' ' Cuba' ' Jamaica' ' India' ' ?' ' Mexico' ' South'
 ' Puerto-Rico' ' Honduras' ' England' ' Canada' ' Germany' ' Iran'
 ' Philippines' ' Italy' ' Poland' ' Columbia' ' Cambodia' ' Thailand'
 ' Ecuador' ' Laos' ' Taiwan' ' Haiti' ' Portugal' ' Dominican-Republic'
 ' El-Salvador' ' France' ' Guatemala' ' China' ' Japan' ' Yugoslavia'
 ' Peru' ' Outlying-US(Guam-USVI-etc)' ' Scotland' ' Trinadad&Tobago'
 ' Greece' ' Nicaragua' ' Vietnam' ' Hong' ' Ireland' ' Hungary'
 ' Holand-Netherlands']
----------------------------------------
Kolom: income
[' <=50K' ' >50K']
----------------------------------------
```

karena ada beberapa record yang memiliki value "?" maka akan kita ganti dengan "NaN" kemudian akan kita ganti dengan data mayoritas untuk data bertipe kategorikal, dan mean untuk data yang bertipe numerik

```
# Class untuk Preprocessing Data
class DataPreprocessor:
    def __init__(self, df):
        self.df = df.copy()
        self.df_cleaned = None
```

```
        self.fill_values = {}

    def clean_whitespace(self):
        self.df = self.df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

    def replace_missing_values(self):
        self.df.replace("?", np.nan, inplace=True)

    def fill_missing_values(self):
        for col in self.df.columns:
            if self.df[col].dtype == "object":
                if not self.df[col].mode().empty:
                    mode_value = self.df[col].mode().iloc[0]
                    self.df[col].fillna(mode_value, inplace=True)
                    self.fill_values[col] = mode_value
            else:
                mean_value = self.df[col].astype(float).mean()
                self.df[col].fillna(mean_value, inplace=True)
                self.fill_values[col] = mean_value

    def remove_rows_with_missing_values(self):
        self.df_cleaned = self.df.dropna()

    def preprocess(self):
        self.clean_whitespace()
        self.replace_missing_values()
        self.fill_missing_values()
        self.remove_rows_with_missing_values()

    def get_processed_data(self):
        return self.df_cleaned if self.df_cleaned is not None else self.df

# Preprocess Data
preprocessor = DataPreprocessor(df)
preprocessor.preprocess()
df = preprocessor.get_processed_data()
```

```
  self.df[col].fillna(mode_value, inplace=True)
<ipython-input-11-42455891a13d>:23: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always beh

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) i


  self.df[col].fillna(mean_value, inplace=True)
<ipython-input-11-42455891a13d>:19: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always beh

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) i


  self.df[col].fillna(mode_value, inplace=True)
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  self.df[col].fillna(mean_value, inplace=True)
<ipython-input-11-42455891a13d>:19: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an
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  self.df[col].fillna(mode_value, inplace=True)
<ipython-input-11-42455891a13d>:23: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an
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  self.df[col].fillna(mean_value, inplace=True)
<ipython-input-11-42455891a13d>:23: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an
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  self.df[col].fillna(mean_value, inplace=True)
<ipython-input-11-42455891a13d>:23: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an
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  self.df[col].fillna(mean_value, inplace=True)
<ipython-input-11-42455891a13d>:19: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always beh

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) i


  self.df[col].fillna(mode_value, inplace=True)
```

```
kolom = ["workclass", "education", "marital_status",    "occupation",   "relationship", "race", "sex", "native_country",    "income"]
for col in kolom:
    print(f"Kolom: {col}")
    print(df[col].unique())
    print("-" * 40)
```

```
Kolom: workclass
['Self-emp-not-inc' 'Private' 'State-gov' 'Federal-gov' 'Local-gov'
 'Self-emp-inc' 'Without-pay' 'Never-worked']
----------------------------------------
Kolom: education
['Bachelors' 'HS-grad' '11th' 'Masters' '9th' 'Some-college' 'Assoc-acdm'
 'Assoc-voc' '7th-8th' 'Doctorate' 'Prof-school' '5th-6th' '10th'
 '1st-4th' 'Preschool' '12th']
----------------------------------------
Kolom: marital_status
['Married-civ-spouse' 'Divorced' 'Married-spouse-absent' 'Never-married'
 'Separated' 'Married-AF-spouse' 'Widowed']
----------------------------------------
Kolom: occupation
['Exec-managerial' 'Handlers-cleaners' 'Prof-specialty' 'Other-service'
 'Adm-clerical' 'Sales' 'Craft-repair' 'Transport-moving'
 'Farming-fishing' 'Machine-op-inspct' 'Tech-support' 'Protective-serv'
 'Armed-Forces' 'Priv-house-serv']
----------------------------------------
Kolom: relationship
['Husband' 'Not-in-family' 'Wife' 'Own-child' 'Unmarried' 'Other-relative']
----------------------------------------
Kolom: race
```

```
['White' 'Black' 'Asian-Pac-Islander' 'Amer-Indian-Eskimo' 'Other']
----------------------------------------
Kolom: sex
['Male' 'Female']
----------------------------------------
Kolom: native_country
['United-States' 'Cuba' 'Jamaica' 'India' 'Mexico' 'South' 'Puerto-Rico'
 'Honduras' 'England' 'Canada' 'Germany' 'Iran' 'Philippines' 'Italy'
 'Poland' 'Columbia' 'Cambodia' 'Thailand' 'Ecuador' 'Laos' 'Taiwan'
 'Haiti' 'Portugal' 'Dominican-Republic' 'El-Salvador' 'France'
 'Guatemala' 'China' 'Japan' 'Yugoslavia' 'Peru'
 'Outlying-US(Guam-USVI-etc)' 'Scotland' 'Trinadad&Tobago' 'Greece'
 'Nicaragua' 'Vietnam' 'Hong' 'Ireland' 'Hungary' 'Holand-Netherlands']
----------------------------------------
Kolom: income
['<=50K' '>50K']
----------------------------------------
```

Karena data dengan tipe kategorikal bentuknya masih berupa object dan belum berupa angka, maka selanjutnya kita perlu untuk mengubah agar tipe dari record yang bertipe object menjadi bertipe angka, kita akan menggunakan label encoder dari pustaka sklearn

```
from sklearn.preprocessing import LabelEncoder
kolom = ["workclass", "education", "marital_status",    "occupation",   "relationship", "race", "sex", "native_country",    "income"]

le = LabelEncoder()

for col in kolom:
    df[col] = le.fit_transform(df[col])
df.head()
```

| | age | workclass | fnlwgt | education | education_num | marital_status | occupation | relationship | race | sex | capital_gain | capital_loss | hours_per_week | na |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 50 | 5 | 83311 | 9 | 13 | 2 | 3 | 0 | 4 | 1 | 0 | 0 | 13 | |
| **1** | 38 | 3 | 215646 | 11 | 9 | 0 | 5 | 1 | 4 | 1 | 0 | 0 | 40 | |
| **2** | 53 | 3 | 234721 | 1 | 7 | 2 | 5 | 0 | 2 | 1 | 0 | 0 | 40 | |
| **3** | 28 | 3 | 338409 | 9 | 13 | 2 | 9 | 5 | 2 | 0 | 0 | 0 | 40 | |
| **4** | 37 | 3 | 284582 | 12 | 14 | 2 | 3 | 5 | 4 | 0 | 0 | 0 | 40 | |

Langkah berikutnya: Buat kode dengan df | Lihat plot yang direkomendasikan | New interactive sheet

untuk kolom target (income) ketika penghasilan <= 50k maka diubah ke angka 0, dan jika penghasilan > 50k diubah ke angka 1

### Melakukan Pengecekan Missing Value

```
missing_values = df.isnull().sum()
print(missing_values[missing_values > 0])
```

```
Series([], dtype: int64)
```

```
df.describe()
```

| | age | workclass | fnlwgt | education | education_num | marital_status | occupation | relationship | race | sex | capital_gai |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **count** | 32560.000000 | 32560.000000 | 3.256000e+04 | 32560.000000 | 32560.000000 | 32560.000000 | 32560.000000 | 32560.000000 | 32560.000000 | 32560.000000 | 32560.00000 |
| **mean** | 38.581634 | 3.094349 | 1.897818e+05 | 10.298249 | 10.080590 | 2.611794 | 6.138943 | 1.446376 | 3.665848 | 0.669195 | 1077.61517 |
| **std** | 13.640642 | 1.107094 | 1.055498e+05 | 3.870317 | 2.572709 | 1.506225 | 3.972623 | 1.606794 | 0.848817 | 0.470510 | 7385.40299 |
| **min** | 17.000000 | 0.000000 | 1.228500e+04 | 0.000000 | 1.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.00000 |
| **25%** | 28.000000 | 3.000000 | 1.178315e+05 | 9.000000 | 9.000000 | 2.000000 | 3.000000 | 0.000000 | 4.000000 | 0.000000 | 0.00000 |
| **50%** | 37.000000 | 3.000000 | 1.783630e+05 | 11.000000 | 10.000000 | 2.000000 | 6.000000 | 1.000000 | 4.000000 | 1.000000 | 0.00000 |
| **75%** | 48.000000 | 3.000000 | 2.370545e+05 | 12.000000 | 12.000000 | 4.000000 | 9.000000 | 3.000000 | 4.000000 | 1.000000 | 0.00000 |
| **max** | 90.000000 | 7.000000 | 1.484705e+06 | 15.000000 | 16.000000 | 6.000000 | 13.000000 | 5.000000 | 4.000000 | 1.000000 | 99999.00000 |

## 3. Pemodelan

di sini kita akan menggunakan model Random Forest, kenapa Random Forest?, karena pada hasil pengujian yang telah saya lakukan, Random Forest yang berhasil mendapatkan akurasi tertinggi dibandingkan dengan model-model lainnya

```
# pisahkan fitur dan target
X = df.drop("income", axis=1)
y = df["income"]
```

```
# bagi data menjadi train-test
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

```
# normalisasi fitur numerik
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
# Fungsi evaluasi model
def evaluate_model(model, X_train, X_test, y_train, y_test):
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred, average='weighted')
    recall = recall_score(y_test, y_pred, average='weighted')
    f1 = f1_score(y_test, y_pred, average='weighted')
    report = classification_report(y_test, y_pred)
    cm = confusion_matrix(y_test, y_pred)
    return accuracy, precision, recall, f1, report, cm
```

di sini kami langsung menggunakan empat sampling methods untuk menguji kira-kira dengan tipe handling data imbalance mana akurasi terbaik akan didapatkan

```
# Handling Data Imbalance (Tambahkan opsi "No Sampling")
sampling_methods = {
```

```
    "No Sampling": None,
    "Undersampling": RandomUnderSampler(random_state=42),
    "Oversampling": RandomOverSampler(random_state=42),
    "SMOTE": SMOTE(random_state=42)
}
```

kami menguji menggunakan model Random Forest dan SVM

```
# model yang akan digunakan
models = {
    "Random Forest": RandomForestClassifier(),
    "SVM": SVC()
}
```

parameter yang dicantumkan di sini adalah parameter terbaik yang telah kami uji sebelumnya dengan hyperparameter tuning, sengaja kami cantumkan yang terbaik karena kalau semua parameter kami cantumkan maka tidak efisien pada waktu eksekusi nantinya

```
# hyperparameter
param_grid = {
    "Random Forest": {
        "n_estimators": [200],
        "max_depth": [15],
        "min_samples_split": [5],
        "min_samples_leaf": [1],
        "max_features": [None]
    },
    "SVM": {
        "C": [1],
        "kernel": ["rbf"],
        "gamma": ["scale"]
    }
}
```

di sini proses pembelajaran dilakukan, proses akan menyimpan data parameter terbaik, akurasi, presisi, dll. setelah itu data akan digunakan untuk membuat conf matriks

```
result = {}
conf_matrices = {}

for sampling_name, sampler in sampling_methods.items():
    print(f"\nEvaluating models with {sampling_name}")
    if sampler:
        X_resampled, y_resampled = sampler.fit_resample(X_train, y_train)
    else:
        X_resampled, y_resampled = X_train, y_train

    for name, model in models.items():
        best_params = {}
        if name in param_grid and param_grid[name]:
            grid_search = GridSearchCV(model, param_grid[name], cv=5, scoring="accuracy", n_jobs=-1)
            grid_search.fit(X_resampled, y_resampled)
            model = grid_search.best_estimator_
            best_params = grid_search.best_params_

        accuracy, precision, recall, f1, report, cm = evaluate_model(model, X_resampled, X_test, y_resampled, y_test)
        result[(sampling_name, name)] = {
            'Best Parameters': best_params,
            'Accuracy': accuracy,
            'Precision': precision,
            'Recall': recall,
            'F1 Score': f1,
            'Classification Report': report
        }
        conf_matrices[(sampling_name, name)] = cm
```

```
Evaluating models with No Sampling

Evaluating models with Undersampling

Evaluating models with Oversampling

Evaluating models with SMOTE
```

di bagian ini, model akan ditampilkan berikut dengan akurasi, dll

```
# Menampilkan hasil terbaik untuk setiap model
for model_name, scores in result.items():
    print(f"Model: {model_name}")
    print(f"Best Parameters: {scores['Best Parameters']}")
    print(f"Accuracy: {scores['Accuracy']}")
    print(f"Precision: {scores['Precision']}")
    print(f"Recall: {scores['Recall']}")
    print(f"F1 Score: {scores['F1 Score']}")
    print(f"Classification Report:\n{scores['Classification Report']}")
    print("=" * 50)
```

```
Recall: 0.8409090909090909
F1 Score: 0.846071255224407
Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.86      0.89      4944
           1       0.64      0.78      0.70      1568

    accuracy                           0.84      6512
   macro avg       0.78      0.82      0.80      6512
weighted avg       0.86      0.84      0.85      6512

==================================================
Model: ('SMOTE', 'SVM')
Best Parameters: {'C': 1, 'gamma': 'scale', 'kernel': 'rbf'}
Accuracy: 0.807585995085995
Precision: 0.8583097078413877
Recall: 0.807585995085995
F1 Score: 0.8190903402013153
Classification Report:
              precision    recall  f1-score   support

           0       0.95      0.79      0.86      4944
           1       0.57      0.87      0.69      1568

    accuracy                           0.81      6512
   macro avg       0.76      0.83      0.77      6512
weighted avg       0.86      0.81      0.82      6512

==================================================
```

dari hasil report dapat kita lihat bahwa untuk model Random Forest berhasil mendapatkan akurasi terbaik secara keseluruhan, dengan detail sebagai berikut:

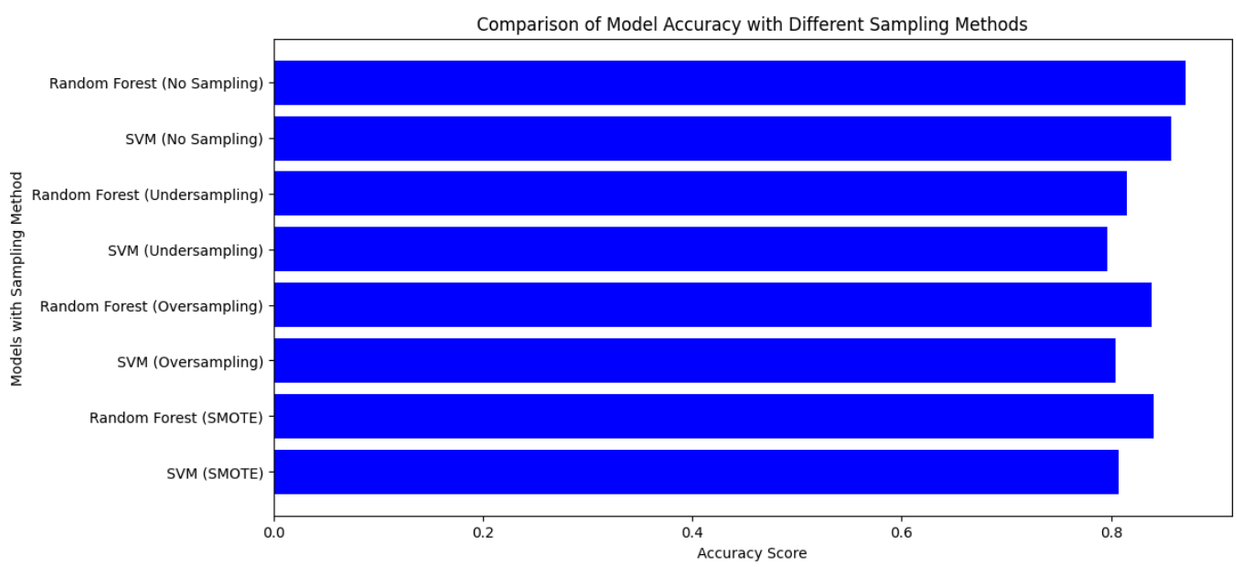
- Random Forest - No Sampling: 0.87 / 87%
- Random Forest - Undersampling: 0.81 / 81%
- Random Forest - Oversampling: 0.84 / 84%
- Random Forest - SMOTE: 0.84 / 84%

sementara itu untuk SVM, akurasi yang didapatkan sebagai berikut:

- SVM - No Sampling: 0.86 / 86%
- SVM - Undersampling: 0.80 / 80%
- SVM - Oversampling: 0.80 / 80%
- SVM - SMOTE: 0.81 / 81%

dari kedua hasil di atas dapat diketahui bahwa akurasi terbaik adalah ketika tidak menggunakan teknik sampling sama sekali (No Sampling)

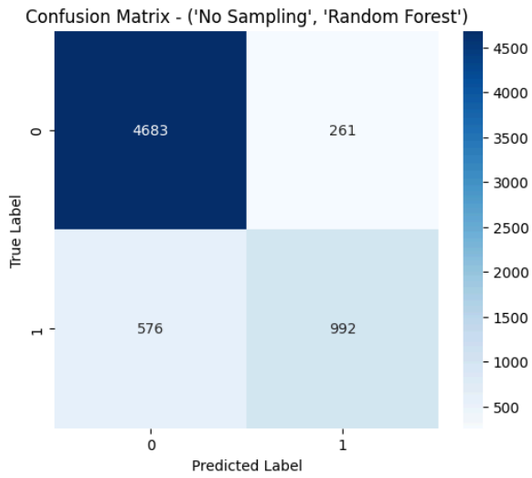
```
# Menampilkan grafik perbandingan akurasi
model_labels = [f"{model} ({sampling})" for sampling, model in result.keys()]
accuracies = [result[key]['Accuracy'] for key in result.keys()]

fig, ax = plt.subplots(figsize=(12, 6))
ax.barh(model_labels, accuracies, color='blue')
ax.set_xlabel("Accuracy Score")
ax.set_ylabel("Models with Sampling Method")
ax.set_title("Comparison of Model Accuracy with Different Sampling Methods")
ax.invert_yaxis()
plt.show()
```

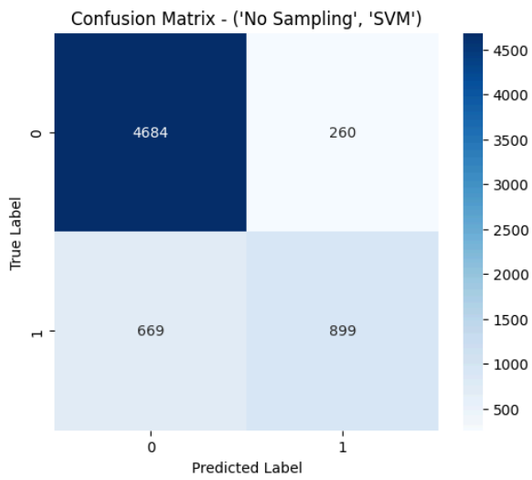


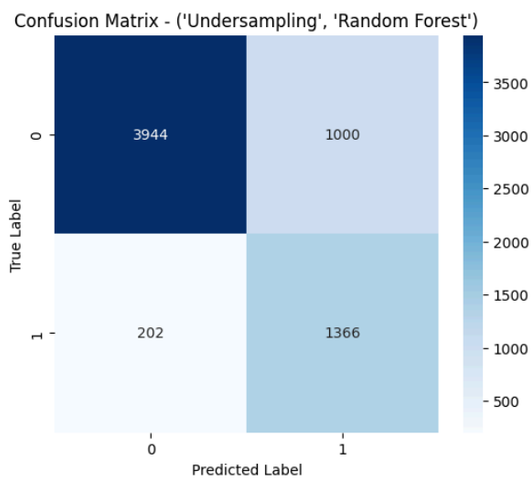

pada visualisasi hasil akurasi dapat kita lihat bahwa model Random Forest memiliki akurasi lebih tinggi dibandingkan dengan model SVM dalam semua keadaan sampling, dan akurasi tertinggi didapatkan model Random Forest dengan teknik samping none (No Sampling)

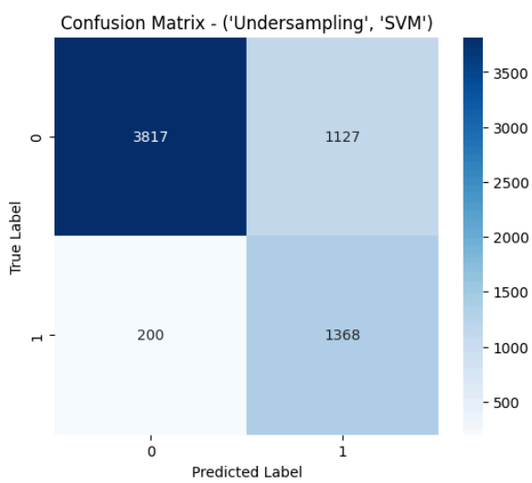
```
# Menampilkan confusion matrix untuk setiap model
for model_name, cm in conf_matrices.items():
    plt.figure(figsize=(6, 5))
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
    plt.xlabel("Predicted Label")
    plt.ylabel("True Label")
    plt.title(f"Confusion Matrix - {model_name}")
    plt.show()
```

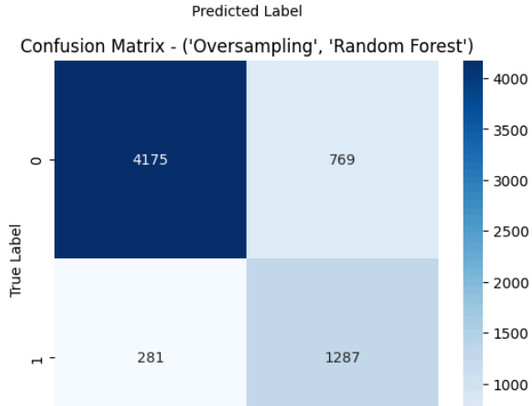
[4]

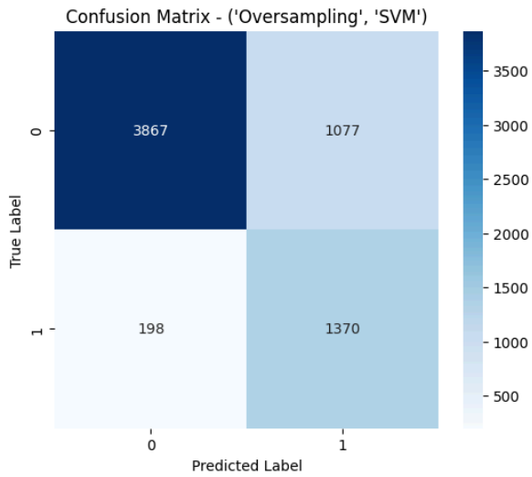

Confusion Matrix - ('No Sampling', 'Random Forest')
4683
261
576
992
True Label
Predicted Label
0
1
4500
4000
3500
3000
2500
2000
1500
1000
500

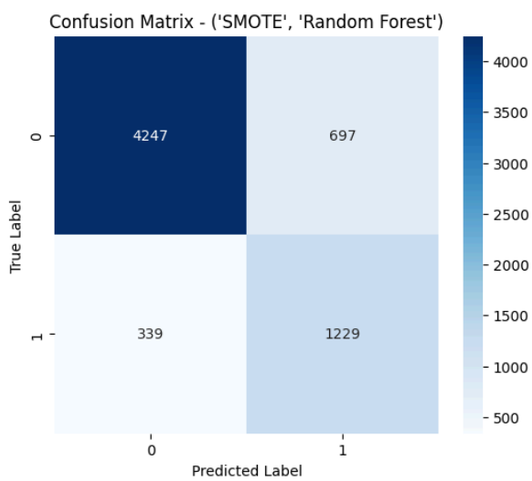

Confusion Matrix - ('No Sampling', 'SVM')
4684
260
669
899
True Label
Predicted Label
0
1
4500
4000
3500
3000
2500
2000
1500
1000
500

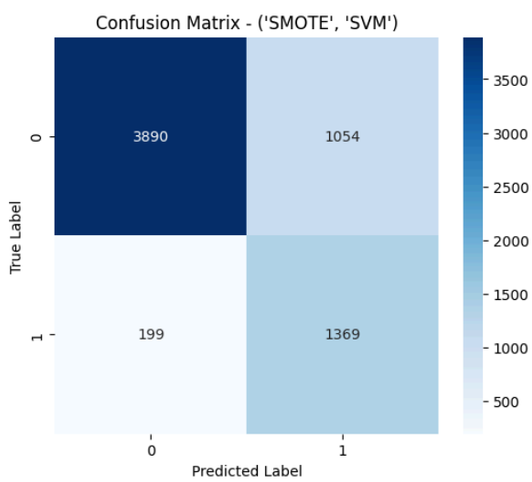

Confusion Matrix - ('Undersampling', 'Random Forest')
3944
1000
202
1366
True Label
Predicted Label
0
1
3500
3000
2500
2000
1500
1000
500


Confusion Matrix - ('Undersampling', 'SVM')
3817
1127
200
1368
True Label
Predicted Label
0
1
3500
3000
2500
2000
1500
1000
500


Confusion Matrix - ('Oversampling', 'Random Forest')
4175
769
281
1287
True Label
0
1
4000
3500
3000
2500
2000
1500
1000




dari hasil conf matrix di atas dapat kita simpulkan bahwa untuk model Random Forest:

1. terdapat 4683 data yang memiliki class negatif dan terklasifikasi sebagai bagian dari class negatif (**Benar**)
2. terdapat 261 data yang memiliki class negatif namun terklasifikasi sebagai bagian dari class positif (**Salah**)
3. terdapat 992 data yang memiliki class positif dan terklasifikasi sebagai bagian dari class positif (**Benar**)
4. terdapat 576 data yang memiliki class positif namun terklasifikasi sebagai bagian dari class positif (**Salah**)

---

sementara itu, untuk model SVM:

1. terdapat 4684 data yang memiliki class negatif dan terklasifikasi sebagai bagian dari class negatif (**Benar**)
2. terdapat 260 data yang memiliki class negatif namun terklasifikasi sebagai bagian dari class positif (**Salah**)
3. terdapat 899 data yang memiliki class positif dan terklasifikasi sebagai bagian dari class positif (**Benar**)
4. terdapat 669 data yang memiliki class positif namun terklasifikasi sebagai bagian dari class positif (**Salah**)

conf matrix yang kami tuliskan di sini mencakup hasil conf matrix dari akurasi terbaik masing-masing model, sehingga hanya kami cantumkan dua saja